2106.07890 — formula report

319 inline formulas, 51 display equations, 0 tables, 6 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2106.07890_EQ0004	007	■ 0.255	`\begin{cases}* \rightarrow * & \text { when } d \text { contains } c, \text { red }, \text { and blue } \\ \emptyset \rightarrow * & \text { when } d \text { does not contain both } c \text { and red, and } d \text { does contain blue } \\ \emptyset \rightarrow \emptyset & \text { when } d \text { does not contain both } c \text { and } \mathrm{red}, \text { and } d \text { does not contain blue. }\end{cases}`	$\begin{cases}* \rightarrow * & \text{when } d \text{ contains } c, \text{ red }, \text{ and blue} \\ \emptyset \rightarrow * & \text{when } d \text{ does not contain both } c \text{ and red, and } d \text{ does contain blue} \\ \emptyset \rightarrow \emptyset & \text{when } d \text{ does not contain both } c \text{ and } \mathrm{red}, \text{ and } d \text{ does not contain blue.}\end{cases}$	$\begin{cases}* \rightarrow * & \text{when } d \text{ contains } c, \textit{red}, \text{and } \textit{blue} \\ \emptyset \rightarrow * & \text{when } d \text{ does not contain both } c \text{ and } \textit{red}, \text{ and } d \text{ does contain } \textit{blue} \\ \emptyset \rightarrow \emptyset & \text{when } d \text{ does not contain both } c \text{ and } \textit{red}, \text{ and } d \text{ does not contain } \textit{blue}.\end{cases}$
2106.07890_EQ0012	011	■ 0.403	`\begin{aligned} \mathcal{L}(\text { red,red firetruck }) & =0.02 \\ \mathcal{L}(\text { red,red idea }) & =10^{-5} \\ \mathcal{L}(\text { red, blue sky }) & =0 \end{aligned}`	$\begin{aligned} \mathcal{L}(\text{ red,red firetruck }) & =0.02 \\ \mathcal{L}(\text{ red,red idea }) & =10^{-5} \\ \mathcal{L}(\text{ red, blue sky }) & =0 \end{aligned}$	$\begin{aligned} \mathcal{L}\left(\textit{red,red firetruck}\right) &= 0.02 \\ \mathcal{L}\left(\textit{red,red idea}\right) &= 10^{-5} \\ \mathcal{L}\left(\textit{red,blue sky}\right) &= 0. \end{aligned}$
2106.07890_EQ0030	018	■ 0.508	`\begin{aligned} \mathcal{L}(c, d) \lim _{W} F(c) & = \min \left\{\frac{\mathcal{L}(c, d) f(c)}{w_{1}}, \frac{\mathcal{L}(c, d) g(c)}{w_{2}}, \mathcal{L}(c, d)\right\} \\ & \leq \min \left\{\frac{f(d)}{w_{1}}, \frac{g(d)}{w_{2}}, 1\right\} \\ & \leq \min \left\{ \frac{f(d)}{w_{1}}, \frac{g(d)}{w_{2}}, 1\right\} \\ & = \lim _{W} F(d) \end{aligned}`	$\begin{aligned} \mathcal{L}(c, d) \lim_{W} F(c) & = \min \left\{\frac{\mathcal{L}(c, d) f(c)}{w_{1}}, \frac{\mathcal{L}(c, d) g(c)}{w_{2}}, \mathcal{L}(c, d)\right\} \\ & \leq \min \left\{\frac{f(d)}{w_{1}}, \frac{g(d)}{w_{2}}, 1\right\} \\ & \leq \min \left\{\frac{f(d)}{w_{1}}, \frac{g(d)}{w_{2}}, 1\right\} \\ & = \lim_{W} F(d) \end{aligned}$	$\begin{aligned} \mathcal{L}(c, d) \lim_{W} F(c) & = \min \left\{\frac{\mathcal{L}(c, d) f(c)}{w_{1}}, \frac{\mathcal{L}(c, d) g(c)}{w_{2}}, \mathcal{L}(c, d)\right\} \\ & \leq \min \left\{\frac{f(d)}{w_{1}}, \frac{g(d)}{w_{2}}, 1\right\} \\ & \leq \min \left\{\frac{f(d)}{w_{1}}, \frac{g(d)}{w_{2}}, 1\right\} \\ & = \lim_{W} F(d), \end{aligned}$
2106.07890_EQ0024 (16)	017	■ 0.538	`\mathcal{E}\left(\operatorname{colim}_{W} F,-\right) \cong \mathcal{V}^{\mathcal{J}^{\mathrm{op}}}(W, \mathcal{E}(F,-))`	$\mathcal{E}\left(\operatorname{colim}_{W} F,-\right) \cong \mathcal{V}^{\mathcal{J}^{\mathrm{op}}}(W, \mathcal{E}(F,-))$	$\mathcal{E}(\operatorname{colim}_{W} F, -) \cong \mathcal{V}^{\mathcal{J}^{\mathrm{op}}}(W, \mathcal{E}(F,-))$

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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2106.07890_EQ0034	019	■0.573	`\begin{aligned} \widehat{\mathcal{L}}(h, f \times g) & = \inf _{c \in \mathcal{L}}\{[h c,(f \times g) c]\} \\ & = \inf _{c \in \mathcal{L}}\{[h c, \min \{f c, g c\}]\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{f c}{h c}, \frac{g c}{h c}, 1\right\} \\ & \left.=\min _{\left\{ \inf _{c \in \mathcal{L}}\right.}\left\{\frac{f c}{h c}, 1\right\}, \inf _{c \in \mathcal{L}}\left\{\frac{g c}{h c}, 1\right\}, 1\right\} \\ & =\widehat{\mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g) \end{aligned}`	$\begin{aligned} \widehat{\mathcal{L}}(h, f \times g) & = \inf_{c \in \mathcal{L}}\{[hc,(f \times g) c]\} \\ & = \inf_{c \in \mathcal{L}}\{[hc, \min \{fc, gc\}]\} \\ & = \inf_{c \in \mathcal{L}}\left\{\frac{fc}{hc}, \frac{gc}{hc}, 1\right\} \\ & = \min_{\{\inf_{c \in \mathcal{L}}} \left\{\frac{fc}{hc}, 1\right\}, \inf_{c \in \mathcal{L}}\left\{\frac{gc}{hc}, 1\right\}, 1\right\} \\ & = \widehat{\mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g) \end{aligned}$	$\begin{aligned} \widehat{\mathcal{L}}(h, f \times g) & = \inf_{c \in \mathcal{L}}\{[hc,(f \times g)c]\} \\ & = \inf_{c \in \mathcal{L}}\{[hc, \min\{fc, gc\}]\} \\ & = \inf_{c \in \mathcal{L}}\left\{\frac{fc}{hc}, \frac{gc}{hc}, 1\right\} \\ & = \min\left\{\inf_{c \in \mathcal{L}}\left\{\frac{fc}{hc}, 1\right\}, \inf_{c \in \mathcal{L}}\left\{\frac{gc}{hc}, 1\right\}, 1\right\} \\ & = \widehat{\mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g) \end{aligned}$
2106.07890_EQ0044	021	■0.657	`\begin{aligned} {\left[h^{x}, h^{y}\right](c) } & = \widehat{\mathcal{L}}\left(h^{c} \times h^{x}, h^{y}\right) \\ & =\inf _{d \in \mathcal{L}}\left\{\left[\left(h^{c} \times h^{x}\right)(d), h^{y}(d)\right]\right\} \\ & =\inf _{d \in \mathcal{L}}\left\{\frac{\pi(d \mid y)}{\pi(d \mid c) \times \pi(d \mid x)}, 1\right\} \\ & =\inf _{d \in \mathcal{L}}\left\{\frac{\pi(d \mid y)}{\min \{\pi(d \mid c), \pi(d \mid x)\}}, 1\right\} . \end{aligned}`	$\begin{aligned} \left[h^{x}, h^{y}\right](c) & = \widehat{\mathcal{L}}\left(h^{c} \times h^{x}, h^{y}\right) \\ & = \inf_{d \in \mathcal{L}}\left\{\left[\left(h^{c} \times h^{x}\right)(d), h^{y}(d)\right]\right\} \\ & = \inf_{d \in \mathcal{L}}\left\{\frac{\pi(d \mid y)}{\pi(d \mid c) \times \pi(d \mid x)}, 1\right\} \\ & = \inf_{d \in \mathcal{L}}\left\{\frac{\pi(d \mid y)}{\min \{\pi(d \mid c), \pi(d \mid x)\}}, 1\right\} . \end{aligned}$	$\begin{aligned} [h^x, h^y](c) & = \widehat{\mathcal{L}}(h^c \times h^x, h^y) \\ & = \inf_{d \in \mathcal{L}}\{[(h^c \times h^x)(d), h^y(d)]\} \\ & = \inf_{d \in \mathcal{L}}\left\{\frac{\pi(d\|y)}{\pi(d\|c) \times \pi(d\|x)}, 1\right\} . \\ & = \inf_{d \in \mathcal{L}}\left\{\frac{\pi(d\|y)}{\min\{\pi(d\|c), \pi(d\|x)\}}, 1\right\} . \end{aligned}$
2106.07890_ EQ0003 (2)	007	■0.667	`\left\{\left(h^{c} \times h^{\text {red }}\right)(d) \rightarrow h^{\text {blue }}(d)\right\}_{d \in \text { ob }(\mathrm{C})}`	$\left\{\left(h^{c} \times h^{\text{red}}\right)(d) \rightarrow h^{\text{blue}}(d)\right\}_{d \in \text{ ob }(\mathrm{C})}$	$\{(h^c \times h^{red})(d) \to h^{blue}(d)\}_{d \in \mathrm{ob}(\mathsf{C})}$
2106.07890_ EQ0020 (12)	014	■0.776	`c \mapsto h^{x}(c):= \begin{cases}\pi(c \mid x) & \text { if } x \leq c \\ 0 & \text { otherwise }\end{cases}`	$c \mapsto h^{x}(c):= \begin{cases}\pi(c \mid x) & \text{ if } x \leq c \\ 0 & \text{ otherwise }\end{cases}$	$c \mapsto h^x(c) := \begin{cases}\pi(c\|x) & \text{if } x \leq c \\ 0 & \text{otherwise.}\end{cases}$
2106.07890_ EQ0032 (19)	018	■0.783	`\left(\left(w_{1}, h^{x}\right) \times\left(w_{2}, h^{y}\right)\right)(c)= \begin{cases}\min \left\{\frac{\pi(c \mid x)}{w_{1}}, \frac{\pi(c \mid y)}{w_{2}}, 1\right\} & \text { if } x \leq c \text { and } y \leq c \\ 0 & \text { otherwise }\end{cases}`	$\left(\left(w_{1}, h^{x}\right) \times\left(w_{2}, h^{y}\right)\right)(c)= \begin{cases}\min \left\{\frac{\pi(c \mid x)}{w_{1}}, \frac{\pi(c \mid y)}{w_{2}}, 1\right\} & \text{ if } x \leq c \text{ and } y \leq c \\ 0 & \text{ otherwise }\end{cases}$	$((w_1, h^x) \times (w_2, h^y))(c) = \begin{cases}\min\left\{\frac{\pi(c\|x)}{w_1}, \frac{\pi(c\|y)}{w_2}, 1\right\} & \text{if } x \leq c \text{ and } y \leq c \\ 0 & \text{otherwise.}\end{cases}$
2106.07890_EQ0005	007	■0.797	`\left[h^{x}, h^{y}\right](c)= \begin{cases}* & \text { if every text that contains both } c \text { and } x \text { also contains } y \\ \emptyset & \text { otherwise }\end{cases}`	$\left[h^{x}, h^{y}\right](c)= \begin{cases}* & \text{ if every text that contains both } c \text{ and } x \text{ also contains } y \\ \emptyset & \text{ otherwise }\end{cases}$	$[h^x, h^y](c) = \begin{cases}* & \text{if every text that contains both } c \text{ and } x \text{ also contains } y \\ \emptyset & \text{otherwise.}\end{cases}$

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2106.07890_ EQ0008 (6)	010	■ 0.890	`[a, b]:= \begin{cases}b / a & \text { if } b<a \\ 1 & \text { otherwise } .\end{cases}`	$[a, b]:= \begin{cases}b / a & \text { if } b<a \\ 1 & \text { otherwise } .\end{cases}$	$[a, b] := \begin{cases} b/a & \textit{if } b<a, \\ 1 & \textit{otherwise}. \end{cases}$
2106.07890_EQ0040	020	■ 0.955	`\begin{aligned} \mathcal{L}(c, d) \operatorname{colim}_{W} F(c) & =\max \left\{ \mathcal{L}(c, d) w_{1} f(c), \mathcal{L}(c, d) w_{2} g(c)\right\} \\ & \leq \max \left\{\frac{f(d)}{w_{1}}, \frac{g(d)}{w_{2}}\right\} \\ & =\operatorname{colim}_{W} F(d) \end{aligned}`	$\begin{aligned} \mathcal{L}(c, d) \operatorname{colim}_{W} F(c) & =\max \left\{ \mathcal{L}(c, d) w_{1} f(c), \mathcal{L}(c, d) w_{2} g(c)\right\} \\ & \leq \max \left\{\frac{f(d)}{w_{1}}, \frac{g(d)}{w_{2}}\right\} \\ & =\operatorname{colim}_{W} F(d) \end{aligned}$	$\begin{aligned} \mathcal{L}(c, d) \operatorname{colim}_{W} F(c) & =\max \left\{ \mathcal{L}(c, d) w_{1} f(c), \mathcal{L}(c, d) w_{2} g(c)\right\} \\ & \leq \max \left\{\frac{f(d)}{w_{1}}, \frac{g(d)}{w_{2}}\right\} \\ & =\operatorname{colim}_{W} F(d), \end{aligned}$
2106.07890_EQ0048	023	■ 0.967	`\begin{aligned} \left(\left(w_{1}, f\right) \times \left(w_{2}, g\right)\right)(c) & =-\ln \left(\left(\left(w_{1}^{\prime}, f^{\prime}\right) \times \left(w_{2}^{\prime}, g^{\prime}\right)\right)(c)\right) \\ & =-\ln \left(\min \left\{\frac{f^{\prime} c}{w_{1}^{\prime}}, \frac{g^{\prime} c}{w_{2}^{\prime}}, 1\right \}\right) \\ & =\max \left\{-\ln \left(\frac{f^{\prime} c}{w_{1}^{\prime}}\right),-\ln \left(\frac{g^{\prime} c}{w_{2}^{\prime}}\right), 0\right\} \\ & =\max \left\{-\ln \left(f^{\prime} c\right)+\ln \left(w_{1}^{\prime} \right),-\ln \left(g^{\prime} c\right)+\ln \left(w_{2}^{\prime}\right), 0\right\} \\ & =\max \left\{f c-w_{1}, g c-w_{2}, 0\right\} \end{aligned}`	$\begin{aligned} \left(\left(w_{1}, f\right) \times \left(w_{2}, g\right)\right)(c) & =-\ln \left(\left(\left(w_{1}^{\prime}, f^{\prime}\right) \times \left(w_{2}^{\prime}, g^{\prime}\right)\right)(c)\right) \\ & =-\ln \left(\min \left\{\frac{f^{\prime} c}{w_{1}^{\prime}}, \frac{g^{\prime} c}{w_{2}^{\prime}}, 1\right \}\right) \\ & =\max \left\{-\ln \left(\frac{f^{\prime} c}{w_{1}^{\prime}}\right),-\ln \left(\frac{g^{\prime} c}{w_{2}^{\prime}}\right), 0\right\} \\ & =\max \left\{-\ln \left(f^{\prime} c\right)+\ln \left(w_{1}^{\prime} \right),-\ln \left(g^{\prime} c\right)+\ln \left(w_{2}^{\prime}\right), 0\right\} \\ & =\max \left\{f c-w_{1}, g c-w_{2}, 0\right\} \end{aligned}$	$\begin{aligned} ((w_1, f) \times (w_2, g))(c) & = -\ln\left(((w_1', f') \times (w_2', g'))(c)\right) \\ & = -\ln\left(\min\left\{\frac{f'c}{w_1'}, \frac{g'c}{w_2'}, 1\right\}\right) \\ & = \max\left\{-\ln\left(\frac{f'c}{w_1'}\right), -\ln\left(\frac{g'c}{w_2'}\right), 0\right\} \\ & = \max\left\{-\ln(f'c) + \ln(w_1'), -\ln(g'c) + \ln(w_2'), 0\right\} \\ & = \max\left\{fc - w_1, gc - w_2, 0\right\}. \end{aligned}$
2106.07890_EQ0029	017	■ 0.990	`\begin{aligned} {[0,1]^{\mathcal{J}}(W, \widehat{ \mathcal{L}}(Z, F)) } & =\min \left\{\left[w_{1}, \inf _{c \in \mathcal{L}}\left\{\frac{f c}{Z c}, 1\right\} \right],\left[w_{2}, \inf _{c \in \mathcal{L}}\left\{ \frac{g c}{Z c}, 1\right\}\right]\right\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{f c}{w_{1} Z c}, \frac{g c}{w_{2} Z c}, 1\right\} . \end{aligned}`	$\begin{aligned} {[0,1]^{\mathcal{J}}(W, \widehat{ \mathcal{L}}(Z, F)) } & =\min \left\{\left[w_{1}, \inf _{c \in \mathcal{L}}\left\{\frac{f c}{Z c}, 1\right\} \right],\left[w_{2}, \inf _{c \in \mathcal{L}}\left\{ \frac{g c}{Z c}, 1\right\}\right]\right\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{f c}{w_{1} Z c}, \frac{g c}{w_{2} Z c}, 1\right\} . \end{aligned}$	$\begin{aligned} [0,1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F)) & = \min\left\{\left[w_1, \inf_{c \in \mathcal{L}}\left\{\frac{fc}{Zc}, 1\right\}\right], \left[w_2, \inf_{c \in \mathcal{L}}\left\{\frac{gc}{Zc}, 1\right\}\right]\right\} \\ & = \inf_{c \in \mathcal{L}}\left\{\frac{fc}{w_1 Zc}, \frac{gc}{w_2 Zc}, 1\right\}. \end{aligned}$

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2106.07890_EQ0037	019	■0.996	`\begin{aligned} \widehat{\mathcal{L}}\left(\operatorname{colim}_{W} F i, Z\right) & =\inf _{c \in \mathcal{L}}\left\{\left[\max \left\{w_{1} f c, w_{2} g c\right\}, Z c\right]\right\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{\max \left\{w_{1} f c, w_{2} g c\right\}}, 1\right\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{w_{1} f c}, \frac{Z c}{w_{2} g c}\right\} . \end{aligned}`	$\begin{aligned} \widehat{\mathcal{L}}\left(\operatorname{colim}_{W} F i, Z\right) & =\inf _{c \in \mathcal{L}}\left\{\left[\max \left\{w_{1} f c, w_{2} g c\right\}, Z c\right]\right\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{\max \left\{w_{1} f c, w_{2} g c\right\}}, 1\right\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{w_{1} f c}, \frac{Z c}{w_{2} g c}\right\} . \end{aligned}$	$\begin{aligned} \widehat{\mathcal{L}}(\operatorname{colim}_{W} Fi, Z) & =\inf _{c \in \mathcal{L}}\{[\max \{w_{1} f c, w_{2} g c\}, Z c]\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{\max \{w_{1} f c, w_{2} g c\}}, 1\right\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{w_{1} f c}, \frac{Z c}{w_{2} g c}\right\} . \end{aligned}$
2106.07890_EQ0039	020	■0.997	`\begin{aligned} {[0,1]^{\mathcal{J}}(W, \widehat{ \mathcal{L}}(F, Z)) } & =\min \left\{\left[w_{1}, \inf _{c \in \mathcal{L}}\left\{\frac{Z c}{f c}, 1\right\} \right],\left[w_{2}, \inf _{c \in \mathcal{L}}\left\{ \frac{Z c}{g c}, 1\right\}\right]\right\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{w_{1} f c}, \frac{Z c}{w_{2} g c}, 1\right\} . \end{aligned}`	$\begin{aligned} {[0,1]^{\mathcal{J}}(W, \widehat{ \mathcal{L}}(F, Z)) } & =\min \left\{\left[w_{1}, \inf _{c \in \mathcal{L}}\left\{\frac{Z c}{f c}, 1\right\} \right],\left[w_{2}, \inf _{c \in \mathcal{L}}\left\{ \frac{Z c}{g c}, 1\right\}\right]\right\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{w_{1} f c}, \frac{Z c}{w_{2} g c}, 1\right\} . \end{aligned}$	$\begin{aligned} [0,1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z)) & =\min \left\{\left[w_{1}, \inf _{c \in \mathcal{L}}\left\{\frac{Z c}{f c}, 1\right\} \right],\left[w_{2}, \inf _{c \in \mathcal{L}}\left\{ \frac{Z c}{g c}, 1\right\}\right]\right\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{w_{1} f c}, \frac{Z c}{w_{2} g c}, 1\right\} . \end{aligned}$
2106.07890_EQ0035	019	■0.999	`\left(w_{1}, f\right) \sqcup\left(w_{2}, g\right):= \operatorname{colim}_{W} F .`	$\left(w_{1}, f\right) \sqcup\left(w_{2}, g\right):= \operatorname{colim}_{W} F .$	$(w_{1}, f) \sqcup (w_{2}, g) := \operatorname{colim}_{W} F.$
2106.07890_EQ0001	007	■1.000	`\left[h^{\text {red }}, h^{\text {blue }}\right]: \mathrm{C} \rightarrow \text { Set }`	$\left[h^{\text {red }}, h^{\text {blue }}\right]: \mathrm{C} \rightarrow \text { Set }$	$[h^{red}, h^{blue}]\colon \mathsf{C} \to \mathsf{Set}$
2106.07890_EQ0010	010	■1.000	`\begin{aligned} & a \times b:=\min \{a, b\} \\ & a \sqcup b:=\max \{a, b\} . \end{aligned}`	$\begin{aligned} & a \times b:=\min \{a, b\} \\ & a \sqcup b:=\max \{a, b\} . \end{aligned}$	$\begin{aligned} a \times b &:= \min\{a, b\} \\ a \sqcup b &:= \max\{a, b\}. \end{aligned}$
2106.07890_ EQ0002 (1)	007	■1.000	`\left[h^{\text {red }}, h^{\text {blue }}\right](c)= \widehat{\mathrm{C}}\left(h^{c},\left[h^{\text {red }}, h^{\text {blue }}\right]\right)=\widehat{\mathrm{C}} \left(h^{c} \times h^{\text {red }}, h^{\text {blue }} \right) .`	$\left[h^{\text {red }}, h^{\text {blue }}\right](c)= \widehat{\mathrm{C}}\left(h^{c},\left[h^{\text {red }}, h^{\text {blue }}\right]\right)=\widehat{\mathrm{C}} \left(h^{c} \times h^{\text {red }}, h^{\text {blue }} \right) .$	$[h^{red}, h^{blue}](c) = \widehat{\mathsf{C}}(h^{c}, [h^{red}, h^{blue}]) = \widehat{\mathsf{C}}(h^{c} \times h^{red}, h^{blue}).$
2106.07890_ EQ0006 (4)	009	■1.000	`\begin{gathered} 1 \leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) \otimes \mathcal{C}(x, y) \leq \mathcal{C}(x, z) \end{gathered}`	$\begin{gathered} 1 \leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) \otimes \mathcal{C}(x, y) \leq \mathcal{C}(x, z) \end{gathered}$	$\begin{gathered} 1 \leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) \otimes \mathcal{C}(x, y) \leq \mathcal{C}(x, z) \end{gathered}$
2106.07890_ EQ0007 (5)	009	■1.000	`x \otimes y \leq z \text { if and only if } x \leq[y, z]`	$x \otimes y \leq z \text { if and only if } x \leq[y, z]$	$x \otimes y \leq z$ if and only if $x \leq [y, z]$
2106.07890_ EQ0009 (7)	010	■1.000	`a b \leq c \text { if and only if } a \leq[b, c]`	$a b \leq c \text { if and only if } a \leq[b, c]$	$ab \leq c$ if and only if $a \leq [b, c]$
2106.07890_EQ0011	011	■1.000	`\mathcal{L}(x, y):=\pi(y \mid x),`	$\mathcal{L}(x, y):=\pi(y \mid x),$	$\mathcal{L}(x, y) := \pi(y \vert x),$
2106.07890_ EQ0013 (8)	011	■1.000	`\pi(z \mid y) \pi(y \mid x)=\pi(z \mid x)`	$\pi(z \mid y) \pi(y \mid x)=\pi(z \mid x)$	$\pi(z \vert y)\pi(y \vert x) = \pi(z \vert x)$
2106.07890_ EQ0014 (9)	012	■1.000	`\mathcal{C}(x, y) \leq \mathcal{D}(f x, f y)`	$\mathcal{C}(x, y) \leq \mathcal{D}(f x, f y)$	$\mathcal{C}(x, y) \leq \mathcal{D}(fx, fy)$

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2106.07890_EQ0015 (10)	012	■ 1.000	`\mathcal{D}^{\mathcal{C}}(f, g):=\int_{c \in \mathcal{C}} \mathcal{D}(f c, g c)`	$\mathcal{D}^{\mathcal{C}}(f, g):=\int_{c \in \mathcal{C}} \mathcal{D}(f c, g c)$	$\mathcal{D}^{\mathcal{C}}(f,g) := \int_{c\in\mathcal{C}} \mathcal{D}(fc, gc)$
2106.07890_EQ0016 (11)	013	■ 1.000	`\widehat{\mathcal{C}}(f, g)=\inf _{c \in \mathcal{C}}\{[f c, g c]\}`	$\widehat{\mathcal{C}}(f, g)=\inf _{c \in \mathcal{C}}\{[f c, g c]\}$	$\widehat{\mathcal{C}}(f,g) = \inf_{c\in\mathcal{C}}\{[fc, gc]\}$
2106.07890_EQ0017	013	■ 1.000	`h^{x}:=\mathcal{C}(x,-)`	$h^{x}:=\mathcal{C}(x,-)$	$h^x := \mathcal{C}(x,-)$
2106.07890_EQ0018	013	■ 1.000	`\mathcal{C}(c, d) \leq\left[h^{x}(c), h^{x}(d)\right]`	$\mathcal{C}(c, d) \leq\left[h^{x}(c), h^{x}(d)\right]$	$\mathcal{C}(c,d) \leq [h^x(c), h^x(d)]$
2106.07890_EQ0019	013	■ 1.000	`\widehat{\mathcal{C}}\left(h^{x}, f\right) \leq \left[h^{x}(x), f x\right]=[1, f x]=f x .`	$\widehat{\mathcal{C}}\left(h^{x}, f\right) \leq \left[h^{x}(x), f x\right]=[1, f x]=f x .$	$\widehat{\mathcal{C}}(h^x, f) \leq [h^x(x), fx] = [1, fx] = fx.$
2106.07890_EQ0021	015	■ 1.000	`x \xrightarrow{a} y \quad h^{y} \xrightarrow{a} h^{x}`	$x \xrightarrow{a} y \quad h^{y} \xrightarrow{a} h^{x}$	$x \xrightarrow{a} y \qquad h^y \xrightarrow{a} h^x$
2106.07890_EQ0022 (13)	016	■ 1.000	`\mathrm{C}(-, \lim F) \cong \operatorname{Set}^{ \mathrm{J}}(*, \mathrm{C}(-, F))`	$\mathrm{C}(-, \lim F) \cong \operatorname{Set}^{ \mathrm{J}}(*, \mathrm{C}(-, F))$	$\mathsf{C}(-, \lim F) \cong \mathsf{Set}^{\mathsf{J}}(*, \mathsf{C}(-, F))$
2106.07890_EQ0023 (15)	016	■ 1.000	`\mathcal{E}\left(-, \lim _{W} F\right) \cong \mathcal{V}^{\mathcal{J}}(W, \mathcal{E}(-, F))`	$\mathcal{E}\left(-, \lim _{W} F\right) \cong \mathcal{V}^{\mathcal{J}}(W, \mathcal{E}(-, F))$	$\mathcal{E}(-, \lim_W F) \cong \mathcal{V}^{\mathcal{J}}(W, \mathcal{E}(-, F))$
2106.07890_EQ0025	017	■ 1.000	`\left(w_{1}, f\right) \times\left(w_{2}, g\right):=\lim _{W} F .`	$\left(w_{1}, f\right) \times\left(w_{2}, g\right):=\lim _{W} F .$	$(w_1, f) \times (w_2, g) := \lim_W F.$
2106.07890_EQ0026 (17)	017	■ 1.000	`\widehat{\mathcal{L}}\left(Z, \lim _{W} F\right) \cong[0,1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F)) .`	$\widehat{\mathcal{L}}\left(Z, \lim _{W} F\right) \cong[0,1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F)) .$	$\widehat{\mathcal{L}}(Z, \lim_W F) \cong [0,1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F)).$
2106.07890_EQ0027	017	■ 1.000	`\inf _{c \in \mathcal{L}}\left\{\left[Z c, \min \left\{ \frac{f c}{w_{1}}, \frac{g c}{w_{2}}, 1\right\}\right] \right\}=\inf _{c \in \mathcal{L}}\left\{\frac{f c}{w_{1} Z c}, \frac{g c}{w_{2} Z c}, 1\right\} .`	$\inf _{c \in \mathcal{L}}\left\{\left[Z c, \min \left\{ \frac{f c}{w_{1}}, \frac{g c}{w_{2}}, 1\right\}\right] \right\}=\inf _{c \in \mathcal{L}}\left\{\frac{f c}{w_{1} Z c}, \frac{g c}{w_{2} Z c}, 1\right\} .$	$\inf_{c\in\mathcal{L}} \left\{ \left[Zc, \min\left\{ \frac{fc}{w_1}, \frac{gc}{w_2}, 1 \right\} \right] \right\} = \inf_{c\in\mathcal{L}} \left\{ \frac{fc}{w_1 Zc}, \frac{gc}{w_2 Zc}, 1 \right\}.$
2106.07890_EQ0028	017	■ 1.000	`\widehat{\mathcal{L}}(Z, F i)=\inf _{c \in \mathcal{L}} \{[Z c, F i c]\}=\inf _{c \in \mathcal{L}}\left\{\frac{F i c}{Z c}, 1\right\} .`	$\widehat{\mathcal{L}}(Z, F i)=\inf _{c \in \mathcal{L}} \{[Z c, F i c]\}=\inf _{c \in \mathcal{L}}\left\{\frac{F i c}{Z c}, 1\right\} .$	$\widehat{\mathcal{L}}(Z, Fi) = \inf_{c\in\mathcal{L}}\{[Zc, Fic]\} = \inf_{c\in\mathcal{L}}\left\{\frac{Fic}{Zc}, 1\right\}.$
2106.07890_EQ0031 (18)	018	■ 1.000	`\left(\left(w_{1}, h^{x}\right) \times\left(w_{2}, h^{y} \right)\right)(c)=\min \left\{\frac{h^{x}(c)}{w_{1}}, \frac{h^{y}(c)}{w_{2}}, 1\right\} .`	$\left(\left(w_{1}, h^{x}\right) \times\left(w_{2}, h^{y} \right)\right)(c)=\min \left\{\frac{h^{x}(c)}{w_{1}}, \frac{h^{y}(c)}{w_{2}}, 1\right\} .$	$((w_1, h^x) \times (w_2, h^y))(c) = \min\left\{\frac{h^x(c)}{w_1}, \frac{h^y(c)}{w_2}, 1\right\}.$
2106.07890_EQ0033	018	■ 1.000	`\widehat{\mathcal{L}}(h, f \times g)=\widehat{ \mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g) .`	$\widehat{\mathcal{L}}(h, f \times g)=\widehat{ \mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g) .$	$\widehat{\mathcal{L}}(h, f\times g) = \widehat{\mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g).$
2106.07890_EQ0036 (20)	019	■ 1.000	`\widehat{\mathcal{L}}\left(\operatorname{colim}_{W} F, Z \right)=[0,1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z)) .`	$\widehat{\mathcal{L}}\left(\operatorname{colim}_{W} F, Z \right)=[0,1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z)) .$	$\widehat{\mathcal{L}}(\mathrm{colim}_W F, Z) = [0,1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z)).$
2106.07890_EQ0038	019	■ 1.000	`\widehat{\mathcal{L}}(F i, Z)=\inf _{c \in \mathcal{L}} \{[F i c, Z c]\}=\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{F i c}, 1\right\} .`	$\widehat{\mathcal{L}}(F i, Z)=\inf _{c \in \mathcal{L}} \{[F i c, Z c]\}=\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{F i c}, 1\right\} .$	$\widehat{\mathcal{L}}(Fi, Z) = \inf_{c\in\mathcal{L}}\{[Fic, Zc]\} = \inf_{c\in\mathcal{L}}\left\{\frac{Zc}{Fic}, 1\right\}.$
2106.07890_EQ0041	020	■ 1.000	`[f, g](c):=\widehat{\mathcal{L}}\left(h^{c} \times f, g \right) .`	$[f, g](c):=\widehat{\mathcal{L}}\left(h^{c} \times f, g \right) .$	$[f,g](c) := \widehat{\mathcal{L}}(h^c \times f, g).$

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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2106.07890_EQ0042	021	■ 1.000	`\widehat{\mathcal{L}}(h \times f, g)=\widehat{ \mathcal{L}}(h,[f, g]) .`	$\widehat{\mathcal{L}}(h \times f, g)=\widehat{\mathcal{L}}(h,[f, g]) .$	$\widehat{\mathcal{L}}(h \times f, g)=\widehat{\mathcal{L}}(h,[f, g]).$
2106.07890_ EQ0043 (21)	021	■ 1.000	`\left[h^{x}, h^{y}\right](c)=\inf _{d \in \mathcal{L}} \left\{\frac{\pi(d \mid y)}{\min \{\pi(d \mid c), \pi(d \mid x)\}}, 1\right\} .`	$\left[h^{x}, h^{y}\right](c)=\inf _{d \in \mathcal{L}} \left\{\frac{\pi(d \mid y)}{\min \{\pi(d \mid c), \pi(d \mid x)\}}, 1\right\} .$	$[h^{x}, h^{y}](c)=\inf_{d \in \mathcal{L}}\left\{\frac{\pi(d\|y)}{\min\{\pi(d\|c), \pi(d\|x)\}}, 1\right\}.$
2106.07890_EQ0045	023	■ 1.000	`\widehat{\mathcal{M}}(f, g)=\sup _{x \in \mathcal{M}} d_{[0, \infty]}(f x, g x)=\sup _{x \in \mathcal{M}} \max \{g x-f x, 0\}`	$\widehat{\mathcal{M}}(f, g)=\sup _{x \in \mathcal{M}} d_{[0, \infty]}(f x, g x)=\sup _{x \in \mathcal{M}} \max \{g x-f x, 0\}$	$\widehat{\mathcal{M}}(f, g)=\sup_{x \in \mathcal{M}} d_{[0, \infty]}(f x, g x)=\sup_{x \in \mathcal{M}} \max\{g x-f x, 0\}$
2106.07890_EQ0046	023	■ 1.000	`\begin{gathered} \left(\left(w_{1}, f\right) \times \left(w_{2}, g\right)\right)(c)=\max \left\{f c-w_{1}, g c-w_{2}, 0\right\} \\ \left(\left(w_{1}, f\right) \sqcup \left(w_{2}, g\right)\right)(c)=\min \left\{f c+w_{1}, g c+w_{2}\right\} \end{gathered}`	$\begin{gathered} \left(\left(w_{1}, f\right) \times \left(w_{2}, g\right)\right)(c)=\max \left\{f c-w_{1}, g c-w_{2}, 0\right\} \\ \left(\left(w_{1}, f\right) \sqcup \left(w_{2}, g\right)\right)(c)=\min \left\{f c+w_{1}, g c+w_{2}\right\} \end{gathered}$	$\begin{gathered} ((w_{1}, f) \times (w_{2}, g))(c)=\max\{f c-w_{1}, g c-w_{2}, 0\} \\ ((w_{1}, f) \sqcup (w_{2}, g))(c)=\min\{f c+w_{1}, g c+w_{2}\} \end{gathered}$
2106.07890_EQ0047	023	■ 1.000	`\left(\left(w_{1}^{\prime}, f^{\prime}\right) \times \left(w_{2}^{\prime}, g^{\prime}\right)\right)(c)=\min \left\{\frac{f^{\prime} c}{w_{1}^{\prime}}, \frac{g^{ \prime} c}{w_{2}^{\prime}}, 1\right\}`	$\left(\left(w_{1}^{\prime}, f^{\prime}\right) \times \left(w_{2}^{\prime}, g^{\prime}\right)\right)(c)=\min \left\{\frac{f^{\prime} c}{w_{1}^{\prime}}, \frac{g^{\prime} c}{w_{2}^{\prime}}, 1\right\}$	$((w_{1}^{\prime}, f^{\prime}) \times (w_{2}^{\prime}, g^{\prime}))(c)=\min\left\{\frac{f^{\prime} c}{w_{1}^{\prime}}, \frac{g^{\prime} c}{w_{2}^{\prime}}, 1\right\}$
2106.07890_EQ0049	024	■ 1.000	`\begin{aligned} \left(\left(w_{1}, f\right) \sqcup \left(w_{2}, g\right)\right)(c) & =-\ln \left(\left(\left(w_{1}^{\prime}, f^{\prime}\right) \sqcup \left(w_{2}^{\prime}, g^{\prime}\right)\right)(c)\right) \\ & =-\ln \left(\max \left\{w_{1}^{\prime} f^{\prime} c, w^{\prime} 2 g^{\prime} c\right\}\right) \\ & =\min \left\{-\ln \left(w_{1}^{\prime} f^{\prime} c\right),-\ln \left(w_{2}^{\prime} g^{\prime} c\right)\right\} \\ & =\min \left\{-\ln \left(f^{\prime} c\right)-\ln \left(w_{1}^{\prime}\right),-\ln \left(g^{\prime} c \right)-\ln \left(w_{2}^{\prime}\right)\right\} \\ & = \min \left\{f c+w_{1}, g c+w_{2}\right\} . \end{aligned}`	$\begin{aligned} \left(\left(w_{1}, f\right) \sqcup \left(w_{2}, g\right)\right)(c) & =-\ln \left(\left(\left(w_{1}^{\prime}, f^{\prime}\right) \sqcup \left(w_{2}^{\prime}, g^{\prime}\right)\right)(c)\right) \\ & =-\ln \left(\max \left\{w_{1}^{\prime} f^{\prime} c, w^{\prime} 2 g^{\prime} c\right\}\right) \\ & =\min \left\{-\ln \left(w_{1}^{\prime} f^{\prime} c\right),-\ln \left(w_{2}^{\prime} g^{\prime} c\right)\right\} \\ & =\min \left\{-\ln \left(f^{\prime} c\right)-\ln \left(w_{1}^{\prime}\right),-\ln \left(g^{\prime} c\right)-\ln \left(w_{2}^{\prime}\right)\right\} \\ & = \min \left\{f c+w_{1}, g c+w_{2}\right\} . \end{aligned}$	$\begin{aligned} ((w_{1}, f) \sqcup (w_{2}, g))(c) & =-\ln\left(((w_{1}^{\prime}, f^{\prime}) \sqcup (w_{2}^{\prime}, g^{\prime}))(c)\right) \\ & =-\ln\left(\max\left\{w_{1}^{\prime} f^{\prime} c, w^{\prime} 2 g^{\prime} c\right\}\right) \\ & =\min\left\{-\ln\left(w_{1}^{\prime} f^{\prime} c\right),-\ln\left(w_{2}^{\prime} g^{\prime} c\right)\right\} \\ & =\min\left\{-\ln(f^{\prime} c)-\ln(w_{1}^{\prime}),-\ln(g^{\prime} c)-\ln(w_{2}^{\prime})\right\} \\ & =\min\left\{f c+w_{1}, g c+w_{2}\right\} . \end{aligned}$
2106.07890_EQ0050	024	■ 1.000	`s_{1} \oplus s_{2}=\min \left\{s_{1}, s_{2}\right\} \text { and } s_{1} \odot s_{2}=s_{1}+s_{2}`	$s_{1} \oplus s_{2}=\min \left\{s_{1}, s_{2}\right\} \text { and } s_{1} \odot s_{2}=s_{1}+s_{2}$	$s_{1} \oplus s_{2}=\min\{s_{1}, s_{2}\}$ and $s_{1} \odot s_{2}=s_{1}+s_{2}$
2106.07890_EQ0051	025	■ 1.000	`\left(\left(w_{1}, f\right) \sqcup\left(w_{2}, g\right) \right)(c)=\min \left\{f c+w_{1}, g c+w_{2}\right\}=w_{1} \odot f c \oplus w_{2} \odot g c .`	$\left(\left(w_{1}, f\right) \sqcup\left(w_{2}, g\right)\right)(c)=\min \left\{f c+w_{1}, g c+w_{2}\right\}=w_{1} \odot f c \oplus w_{2} \odot g c .$	$((w_{1}, f) \sqcup (w_{2}, g))(c)=\min\{f c+w_{1}, g c+w_{2}\}=w_{1} \odot f c \oplus w_{2} \odot g c.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value					

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered
2106.07890_F00001	003	—	`\mathcal{L}`	$\mathcal{L}$
2106.07890_F00002	004	—	`\widehat{\mathcal{L}}`	$\widehat{\mathcal{L}}$
2106.07890_F00003	002	—	`{ }^{+}`	${ }^{+}$
2106.07890_F00004	002	—	`\mathrm{B}^{+}`	B^{+}
2106.07890_F00005	003	—	`\infty`	∞
2106.07890_F00006	005	—	`x \rightarrow y`	$x \rightarrow y$
2106.07890_F00007	005	—	`x`	x
2106.07890_F00008	005	—	`y`	y
2106.07890_F00009	005	—	`z`	z
2106.07890_F00010	005	—	`h^{x}:=\operatorname{hom}(x,-)`	$h^{x}:=\operatorname{hom}(x,-)$
2106.07890_F00011	005	—	`*`	$*$
2106.07890_F00012	005	—	`h^{x}`	h^{x}
2106.07890_F00013	005	—	`\widehat{\mathrm{C}}:=`	$\widehat{\mathrm{C}}:=$
2106.07890_F00014	005	—	`^{\mathrm{C}}`	$^{\mathrm{C}}$
2106.07890_F00015	006	—	`h^{x} \sqcup h^{y}`	$h^{x} \sqcup h^{y}$
2106.07890_F00016	006	—	`x=`	$x=$
2106.07890_F00017	006	—	`y=`	$y=$
2106.07890_F00018	006	—	`h^{\text {red }} \sqcup h^{\text {blue }}`	$h^{\text {red }} \sqcup h^{\text {blue }}$
2106.07890_F00019	006	—	`c`	c
2106.07890_F00020	006	—	`h^{\text {red }}(c) \sqcup h^{\text {blue }}(c)`	$h^{\text {red }}(c) \sqcup h^{\text {blue }}(c)$
2106.07890_F00021	006	—	`h^{x} \times h^{y}: \mathrm{C} \rightarrow`	$h^{x} \times h^{y}: \mathrm{C} \rightarrow$
2106.07890_F00022	006	—	`h^{\text {red }}(c) \times h^{\text {blue }}(c)`	$h^{\text {red }}(c) \times h^{\text {blue }}(c)$
2106.07890_F00023	006	—	`h^{\text {red }} \times h^{\text {blue }}`	$h^{\text {red }} \times h^{\text {blue }}$
2106.07890_F00024	006	—	`\widehat{\mathrm{C}}`	$\widehat{\mathrm{C}}$
2106.07890_F00025	006	—	`[]:, \widehat{\mathrm{C}} \times \widehat{\mathrm{C}} \rightarrow \widehat{\mathrm{C}}`	$[]:, \widehat{\mathrm{C}} \times \widehat{\mathrm{C}} \rightarrow \widehat{\mathrm{C}}$
2106.07890_F00026	006	—	`\widehat{\mathrm{C}}(F \times`	$\widehat{\mathrm{C}}(F \times$
2106.07890_F00027	006	—	`G, H) \cong \widehat{\mathrm{C}}(F,[G, H])`	$G, H) \cong \widehat{\mathrm{C}}(F,[G, H])$
2106.07890_F00028	006	—	`F, G`	F, G
2106.07890_F00029	006	—	`H`	H
2106.07890_F00030	006	—	`\mathrm{A}(x, y)`	$\mathrm{A}(x, y)$
2106.07890_F00031	006	—	`[G, H]`	$[G, H]$
2106.07890_F00032	006	—	`x \leq y`	$x \leq y$
2106.07890_F00033	006	—	`\wedge`	$\wedge$
2106.07890_F00034	006	—	`x \wedge y \leq z`	$x \wedge y \leq z$
2106.07890_F00035	006	—	`x \leq(y \Rightarrow z)`	$x \leq(y \Rightarrow z)$
2106.07890_F00036	006	—	`x, y, z`	x, y, z
2106.07890_F00037	007	—	`h^{c} \times h^{\text {red }}`	$h^{c} \times h^{\text {red }}$
2106.07890_F00038	007	—	`h^{\text {blue }}`	$h^{\text {blue }}$
2106.07890_F00039	007	—	`\left(h^{c} \times h^{\text {red }}\right)(d)`	$\left(h^{c} \times h^{\text {red }}\right)(d)$
2106.07890_F00040	007	—	`\left(h^{\text {blue }}\right)(d)`	$\left(h^{\text {blue }}\right)(d)$
2106.07890_F00041	007	—	`\left[h^{\text {red }}, h^{\text {blue }}\right](c)`	$\left[h^{\text {red }}, h^{\text {blue }}\right](c)$
2106.07890_F00042	007	—	`\emptyset`	$\emptyset$
2106.07890_F00043	007	—	`h^{c} \times`	$h^{c} \times$
2106.07890_F00044	007	—	`h^{\text {red }}(d)=h^{c}(d) \times h^{\text {red }}(d)`	$h^{\text {red }}(d)=h^{c}(d) \times h^{\text {red }}(d)$
2106.07890_F00045	007	—	`d`	d
2106.07890_F00046	007	—	`h^{\text {blue }}(d)`	$h^{\text {blue }}(d)$
2106.07890_F00047	007	—	`\left(h^{c} \times h^{\text {red }}\right)(d)=*`	$\left(h^{c} \times h^{\text {red }}\right)(d)=*$
2106.07890_F00048	007	—	`h^{\text {blue }}(d)=\emptyset`	$h^{\text {blue }}(d)=\emptyset$
2106.07890_F00049	007	—	`\left(h^{c} \times h^{\text {red }}\right)(d) \rightarrow h^{\text {blue }}(d)`	$\left(h^{c} \times h^{\text {red }}\right)(d) \rightarrow h^{\text {blue }}(d)$
2106.07890_F00050	007	—	`\left[h^{\text {red }}, h^{\text {blue }}\right]`	$\left[h^{\text {red }}, h^{\text {blue }}\right]$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Identifier	Page	Conf.	LaTeX source	Rendered
2106.07890_F00051	007	—	`=*`	$=*$
2106.07890_F00052	007	—	`\left[h^{\text {red }}, h^{\text {blue }}\right](`	$\left[h^{\text {red }}, h^{\text {blue }}\right]($
2106.07890_F00053	007	—	`)=\emptyset`	$)=\emptyset$
2106.07890_F00054	007	—	`\left[h^{x}, h^{y}\right]: \mathrm{C} \rightarrow`	$\left[h^{x}, h^{y}\right]: \mathrm{C} \rightarrow$
2106.07890_F00055	008	—	`\mathrm{C}^{\mathrm{op}} \rightarrow`	$\mathrm{C}^{\mathrm{op}} \rightarrow$
2106.07890_F00056	008	—	`h^{x}=\operatorname{hom}(x,-)`	$h^{x}=\operatorname{hom}(x,-)$
2106.07890_F00057	008	—	`\mathrm{C}^{\mathrm{op}}`	C^{op}
2106.07890_F00058	008	—	`\mathrm{C}^{\mathrm{op}}(x, y):=\mathrm{C}(y, x)`	$\mathrm{C}^{\mathrm{op}}(x, y):=\mathrm{C}(y, x)$
2106.07890_F00059	008	—	`\mathrm{C}(x, y)`	$\mathrm{C}(x, y)$
2106.07890_F00060	008	—	`[0,1]`	$[0,1]$
2106.07890_F00061	008	—	`a`	a
2106.07890_F00062	008	—	`b`	b
2106.07890_F00063	009	—	`(\mathcal{V}, \leq, \otimes, 1)`	$(\mathcal{V}, \leq, \otimes, 1)$
2106.07890_F00064	009	—	`(\mathcal{V}, \leq)`	$(\mathcal{V}, \leq)$
2106.07890_F00065	009	—	`(\mathcal{V}, \otimes, 1)`	$(\mathcal{V}, \otimes, 1)$
2106.07890_F00066	009	—	`x \otimes y \leq x^{\prime} \otimes y^{\prime}`	$x \otimes y \leq x^{\prime} \otimes y^{\prime}$
2106.07890_F00067	009	—	`x \leq x^{\prime}`	$x \leq x^{\prime}$
2106.07890_F00068	009	—	`y \leq y^{\prime}`	$y \leq y^{\prime}$
2106.07890_F00069	009	—	`\mathcal{V}, \leq, \otimes, 1`	$\mathcal{V}, \leq, \otimes, 1$
2106.07890_F00070	009	—	`\mathcal{V}`	$\mathcal{V}$
2106.07890_F00071	009	—	`\mathcal{C}`	$\mathcal{C}$
2106.07890_F00072	009	—	`\mathcal{C}(x, y) \in \mathcal{V}`	$\mathcal{C}(x, y) \in \mathcal{V}$
2106.07890_F00073	009	—	`x, y, z \in \mathcal{C}`	$x, y, z \in \mathcal{C}$
2106.07890_F00074	009	—	`[0,1]:=\{x \in \mathbb{R}: 0 \leq x \leq 1\}`	$[0,1]:=\{x \in \mathbb{R}: 0 \leq x \leq 1\}$
2106.07890_F00075	009	—	`\leq`	$\leq$
2106.07890_F00076	009	—	`(x, y) \mapsto \mathcal{C}(x, y)`	$(x, y) \mapsto \mathcal{C}(x, y)$
2106.07890_F00077	009	—	`x, y \in \mathcal{C}`	$x, y \in \mathcal{C}$
2106.07890_F00078	009	—	`\mathcal{C}(x, x)=1`	$\mathcal{C}(x, x)=1$
2106.07890_F00079	009	—	`x \in \mathcal{C}`	$x \in \mathcal{C}$
2106.07890_F00080	009	—	`\mathcal{C}(y, z) \mathcal{C}(x, y) \leq \mathcal{C}(x, z)`	$\mathcal{C}(y, z) \mathcal{C}(x, y) \leq \mathcal{C}(x, z)$
2106.07890_F00081	009	—	`[x, y] \in \mathcal{V}`	$[x, y] \in \mathcal{V}$
2106.07890_F00082	009	—	`\mathcal{V}(x, y)`	$\mathcal{V}(x, y)$
2106.07890_F00083	010	—	`[x, y]`	$[x, y]$
2106.07890_F00084	010	—	`(x, y) \mapsto[x, y]`	$(x, y) \mapsto[x, y]$
2106.07890_F00085	010	—	`1 \leq[x, x]`	$1 \leq[x, x]$
2106.07890_F00086	010	—	`[y, z] \otimes[x, y] \leq[x, z]`	$[y, z] \otimes[x, y] \leq[x, z]$
2106.07890_F00087	010	—	`1 \otimes x \leq x`	$1 \otimes x \leq x$
2106.07890_F00088	010	—	`[y, z] \leq[y, z]`	$[y, z] \leq[y, z]$
2106.07890_F00089	010	—	`[y, z] \otimes y \leq z`	$[y, z] \otimes y \leq z$
2106.07890_F00090	010	—	`[x, y] \otimes x \leq y`	$[x, y] \otimes x \leq y$
2106.07890_F00091	010	—	`[y, z] \otimes[x, y] \otimes x \leq z`	$[y, z] \otimes[x, y] \otimes x \leq z$
2106.07890_F00092	010	—	`a \otimes b:=a b`	$a \otimes b:=a b$
2106.07890_F00093	010	—	`a, b \in[0,1]`	$a, b \in[0,1]$
2106.07890_F00094	010	—	`a b \leq c`	$a b \leq c$
2106.07890_F00095	010	—	`a \leq[b, c]`	$a \leq[b, c]$
2106.07890_F00096	010	—	`c<b`	$c<b$
2106.07890_F00097	010	—	`a \leq \frac{c}{b}=[b, c]`	$a \leq \frac{c}{b}=[b, c]$
2106.07890_F00098	010	—	`b \leq c`	$b \leq c$
2106.07890_F00099	010	—	`[b, c]=1`	$[b, c]=1$
2106.07890_F00100		—	`\square`	$\square$
2106.07890_F00101	010	—	`a, b, c \in[0,1]`	$a, b, c \in[0,1]$
2106.07890_F00102	010	—	`\min \{b / a, 1\}`	$\min \{b / a, 1\}$
2106.07890_F00103	010	—	`b / 0 \geq 1`	$b / 0 \geq 1$
2106.07890_F00104	010	—	`0 \leq b \leq 1`	$0 \leq b \leq 1$
2106.07890_F00105	010	—	`a b`	$a b$
2106.07890_F00106	010	—	`a \otimes b`	$a \otimes b$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Identifier	Page	Conf.	LaTeX source	Rendered
2106.07890_F00107	010	—	`a \times b`	$a \times b$
2106.07890_F00108	010	—	`\min \{a, b\}`	$\min \{a, b\}$
2106.07890_F00109	010	—	`a \sqcup b`	$a \sqcup b$
2106.07890_F00110	010	—	`\mathrm{C}(x, y)=1`	$\mathrm{C}(x, y)=1$
2106.07890_F00111	010	—	`\mathrm{C}(x, y)=0`	$\mathrm{C}(x, y)=0$
2106.07890_F00112	010	—	`2:=\{0,1\}`	$2:=\{0,1\}$
2106.07890_F00113	011	—	`\pi(y \mid x)`	$\pi(y \mid x)$
2106.07890_F00114	011	—	`\pi(y \mid x)=0`	$\pi(y \mid x)=0$
2106.07890_F00115	011	—	`\pi(x \mid x)=`	$\pi(x \mid x)=$
2106.07890_F00116	011	—	`\mathcal{D}`	$\mathcal{D}$
2106.07890_F00117	012	—	`\mathcal{L} \rightarrow \widehat{\mathcal{L}}`	$\mathcal{L} \rightarrow \widehat{\mathcal{L}}$
2106.07890_F00118	012	—	`\mathcal{L} \rightarrow \mathcal{D}`	$\mathcal{L} \rightarrow \mathcal{D}$
2106.07890_F00119	012	—	`\mathcal{D} \cdots \cdots>\widehat{\mathcal{L}}`	$\mathcal{D} \cdots \cdots>\widehat{\mathcal{L}}$
2106.07890_F00120	012	—	`(\mathcal{V}, \otimes, \leq, 1)`	$(\mathcal{V}, \otimes, \leq, 1)$
2106.07890_F00121	012	—	`\mathcal{C} \rightarrow \mathcal{D}`	$\mathcal{C} \rightarrow \mathcal{D}$
2106.07890_F00122	012	—	`f: \mathcal{C} \rightarrow \mathcal{D}`	$f: \mathcal{C} \rightarrow \mathcal{D}$
2106.07890_F00123	012	—	`\mathcal{D}=\mathcal{V}`	$\mathcal{D}=\mathcal{V}$
2106.07890_F00124	012	—	`f: \mathcal{C} \rightarrow \mathcal{V}`	$f: \mathcal{C} \rightarrow \mathcal{V}$
2106.07890_F00125	012	—	`\mathcal{C}(x, y) \leq`	$\mathcal{C}(x, y) \leq$
2106.07890_F00126	012	—	`\mathcal{V}(f x, f y)`	$\mathcal{V}(f x, f y)$
2106.07890_F00127	012	—	`\mathcal{D}^{\mathcal{C}}`	$\mathcal{D}^{\mathcal{C}}$
2106.07890_F00128	012	—	`\mathcal{D}^{\mathcal{C}}(f, g) \in \mathcal{V}`	$\mathcal{D}^{\mathcal{C}}(f, g) \in \mathcal{V}$
2106.07890_F00129	012	—	`f, g: \mathcal{C} \rightarrow \mathcal{D}`	$f, g: \mathcal{C} \rightarrow \mathcal{D}$
2106.07890_F00130	013	—	`\widehat{\mathcal{C}}:=[0,1]^{\mathcal{C}}`	$\widehat{\mathcal{C}}:=[0,1]^{\mathcal{C}}$
2106.07890_F00131	013	—	`f, g: \mathcal{C} \rightarrow[0,1]`	$f, g: \mathcal{C} \rightarrow[0,1]$
2106.07890_F00132	013	—	`\widehat{\mathcal{C}}(f, g)=\inf _{c \in \mathcal{C}}\{1, g c / f c\}`	$\widehat{\mathcal{C}}(f, g)=\inf _{c \in \mathcal{C}}\{1, g c / f c\}$
2106.07890_F00133	013	—	`\mathcal{V}=[0,1]`	$\mathcal{V}=[0,1]$
2106.07890_F00134	013	—	`\mathcal{D}=[0,1]`	$\mathcal{D}=[0,1]$
2106.07890_F00135	013	—	`\mathcal{C} \rightarrow[0,1]`	$\mathcal{C} \rightarrow[0,1]$
2106.07890_F00136	013	—	`c, d \in \mathcal{C}`	$c, d \in \mathcal{C}$
2106.07890_F00137	013	—	`\mathcal{C}(c, d) \leq[\mathcal{C}(x, c), \mathcal{C}(x, d)]`	$\mathcal{C}(c, d) \leq[\mathcal{C}(x, c), \mathcal{C}(x, d)]$
2106.07890_F00138	013	—	`\mathcal{C}(c, d) \mathcal{C}(x, c) \leq \mathcal{C}(x, d)`	$\mathcal{C}(c, d) \mathcal{C}(x, c) \leq \mathcal{C}(x, d)$
2106.07890_F00139	013	—	`h^{x}:=`	$h^{x}:=$
2106.07890_F00140	013	—	`\mathcal{C}(x,-)`	$\mathcal{C}(x,-)$
2106.07890_F00141	013	—	`f: \mathcal{C} \rightarrow[0,1]`	$f: \mathcal{C} \rightarrow[0,1]$
2106.07890_F00142	013	—	`f=h^{x}`	$f=h^{x}$
2106.07890_F00143	013	—	`x \mapsto h^{x}`	$x \mapsto h^{x}$
2106.07890_F00144	013	—	`\widehat{\mathcal{C}}\left(h^{x}, f\right)=f(x)`	$\widehat{\mathcal{C}}\left(h^{x}, f\right)=f(x)$
2106.07890_F00145	013	—	`f`	f
2106.07890_F00146	013	—	`\widehat{\mathcal{C}}\left(h^{x}, f\right)=\inf _{c \in \mathcal{C}}\left\{\left[h^{x}(c), f c \right]\right\}`	$\widehat{\mathcal{C}}\left(h^{x}, f\right)=\inf _{c \in \mathcal{C}}\left\{\left[h^{x}(c), f c\right]\right\}$
2106.07890_F00147	013	—	`c \in \mathcal{C}`	$c \in \mathcal{C}$
2106.07890_F00148	013	—	`\widehat{\mathcal{C}}\left(h^{x}, f\right) \leq\left[h^{x}(c), f c\right]`	$\widehat{\mathcal{C}}\left(h^{x}, f\right) \leq\left[h^{x}(c), f c\right]$
2106.07890_F00149	013	—	`c=x`	$c=x$
2106.07890_F00150	014	—	`\mathcal{C}(x, c) \leq`	$\mathcal{C}(x, c) \leq$
2106.07890_F00151	014	—	`[f x, f c]`	$[f x, f c]$
2106.07890_F00152	014	—	`\mathcal{C}(x, c) \leq[f x, f c]`	$\mathcal{C}(x, c) \leq[f x, f c]$
2106.07890_F00153	014	—	`\mathcal{C}(x, c) f x \leq f c`	$\mathcal{C}(x, c) f x \leq f c$
2106.07890_F00154	014	—	`f x \leq[\mathcal{C}(x, c), f c]`	$f x \leq[\mathcal{C}(x, c), f c]$
2106.07890_F00155	014	—	`f x \leq \inf _{c \in \mathcal{C}}\left\{\left[h^{x}(c), f c\right]\right\}=`	$f x \leq \inf _{c \in \mathcal{C}}\left\{\left[h^{x}(c), f c\right]\right\}=$
2106.07890_F00156	014	—	`\widehat{\mathcal{C}}\left(h^{x}, f\right)`	$\widehat{\mathcal{C}}\left(h^{x}, f\right)$
2106.07890_F00157	014	—	`\mathcal{C}(y, x)=\widehat{\mathcal{C}}\left(h^{x}, h^{y}\right)`	$\mathcal{C}(y, x)=\widehat{\mathcal{C}}\left(h^{x}, h^{y}\right)$
2106.07890_F00158	014	—	`x, y`	x, y
2106.07890_F00159	014	—	`f=h^{y}`	$f=h^{y}$
2106.07890_F00160	014	—	`\widehat{\mathcal{C}}\left(h^{x}, h^{y}\right)=h^{y}(x)=\mathcal{C}(y, x)`	$\widehat{\mathcal{C}}\left(h^{x}, h^{y}\right)=h^{y}(x)=\mathcal{C}(y, x)$

Conf. MathPix confidence: ■ $\geq$0.9 ■ 0.5–0.9 ■ <0.5 — no value

Identifier	Page	Conf.	LaTeX source	Rendered
2106.07890_F00161	014	—	\mathcal{C}^{o p} \rightarrow \widehat{\mathcal{C}}	$\mathcal{C}^{o p} \rightarrow \widehat{\mathcal{C}}$
2106.07890_F00162	014	—	\mathcal{C}^{o p}	$\mathcal{C}^{o p}$
2106.07890_F00163	014	—	\widehat{\mathcal{C}}	$\widehat{\mathcal{C}}$
2106.07890_F00164	014	—	o p	$o p$
2106.07890_F00165	014	—	\mathcal{C}^{\mathrm{op}}	$\mathcal{C}^{\mathrm{op}}$
2106.07890_F00166	014	—	Y	Y
2106.07890_F00167	014	—	\{X \rightarrow Y\}	$\{X \rightarrow Y\}$
2106.07890_F00168	014	—	X	X
2106.07890_F00169	014	—	\widehat{\mathcal{C}}=[0,1]^{\mathcal{C}}	$\widehat{\mathcal{C}}=[0,1]^{\mathcal{C}}$
2106.07890_F00170	014	—	\widehat{\mathcal{L}}:=[0,1]^{\mathcal{L}}	$\widehat{\mathcal{L}}:=[0,1]^{\mathcal{L}}$
2106.07890_F00171	014	—	h^{x}:=\mathcal{L}(x,-)	$h^{x}:=\mathcal{L}(x,-)$
2106.07890_F00172	014	—	x \leq c	$x \leq c$
2106.07890_F00173	015	—	\mathcal{L}(x, y)=a \neq 0	$\mathcal{L}(x, y)=a \neq 0$
2106.07890_F00174	015	—	h^{y}	h^{y}
2106.07890_F00175	015	—	{ }^{\mathrm{C}}	${ }^{\mathrm{C}}$
2106.07890_F00176	016	—	F: \mathrm{J} \rightarrow \mathrm{C}	$F: \mathrm{J} \rightarrow \mathrm{C}$
2106.07890_F00177	016	—	\lim F	$\lim F$
2106.07890_F00178	016	—	Z	Z
2106.07890_F00179	016	—	\mathrm{C}(Z, \lim F) \cong \operatorname{Set}^{\mathrm{J}}(*, \mathrm{C}(Z, F))	$\mathrm{C}(Z, \lim F) \cong \operatorname{Set}^{\mathrm{J}}(*, \mathrm{C}(Z, F))$
2106.07890_F00180	016	—	(Z, F)	(Z, F)
2106.07890_F00181	016	—	*=*(i) \rightarrow \mathrm{C}(Z, F i)	$*=*(i) \rightarrow \mathrm{C}(Z, F i)$
2106.07890_F00182	016	—	i	i
2106.07890_F00183	016	—	Z \rightarrow F i	$Z \rightarrow F i$
2106.07890_F00184	016	—	W: \mathcal{J} \rightarrow \mathcal{V}	$W: \mathcal{J} \rightarrow \mathcal{V}$
2106.07890_F00185	016	—	\mathcal{J}	$\mathcal{J}$
2106.07890_F00186	016	—	\mathcal{E}	$\mathcal{E}$
2106.07890_F00187	016	—	F: \mathcal{J} \rightarrow \mathcal{E}	$F: \mathcal{J} \rightarrow \mathcal{E}$
2106.07890_F00188	016	—	F	F
2106.07890_F00189	016	—	\lim _{W} F	$\lim _{W} F$
2106.07890_F00190	016	—	\mathcal{E} \rightarrow \mathcal{V}	$\mathcal{E} \rightarrow \mathcal{V}$
2106.07890_F00191	017	—	\mathcal{E}\left(Z, \lim _{W} F\right) \cong	$\mathcal{E}\left(Z, \lim _{W} F\right) \cong$
2106.07890_F00192	017	—	\mathcal{V}^{\mathcal{J}}(W, \mathcal{E}(Z, F))	$\mathcal{V}^{\mathcal{J}}(W, \mathcal{E}(Z, F))$
2106.07890_F00193	017	—	W: \mathcal{J}^{\mathrm{op}} \rightarrow \mathcal{V}	$W: \mathcal{J}^{\mathrm{op}} \rightarrow \mathcal{V}$
2106.07890_F00194	017	—	\operatorname{colim}_{W} F	$\operatorname{colim}_{W} F$
2106.07890_F00195	017	—	\mathcal{E}=\widehat{\mathcal{L}}:=[0,1]^{\mathcal{L}}	$\mathcal{E}=\widehat{\mathcal{L}}:=[0,1]^{\mathcal{L}}$
2106.07890_F00196	017	—	\mathcal{J}(i, j)=\delta_{i j}	$\mathcal{J}(i, j)=\delta_{i j}$
2106.07890_F00197	017	—	i, j \in\{1,2\}	$i, j \in\{1,2\}$
2106.07890_F00198	017	—	W: \mathcal{J} \rightarrow	$W: \mathcal{J} \rightarrow$
2106.07890_F00199	017	—	w_{1}:=W(1)	$w_{1}:=W(1)$
2106.07890_F00200	017	—	w_{2}:=	$w_{2}:=$
2106.07890_F00201	017	—	W(2)	$W(2)$
2106.07890_F00202	017	—	f, g: \mathcal{L} \rightarrow[0,1]	$f, g: \mathcal{L} \rightarrow[0,1]$
2106.07890_F00203	017	—	F: \mathcal{J} \rightarrow	$F: \mathcal{J} \rightarrow$
2106.07890_F00204	017	—	f:=F(1)	$f:=F(1)$
2106.07890_F00205	017	—	g:=F(2)	$g:=F(2)$
2106.07890_F00206	017	—	W	W
2106.07890_F00207	017	—	\left(w_{1}, f\right) \times\left(w_{2}, g\right): \mathcal{L} \rightarrow[0,1]	$\left(w_{1}, f\right) \times\left(w_{2}, g\right): \mathcal{L} \rightarrow[0,1]$
2106.07890_F00208	017	—	c \mapsto	$c \mapsto$
2106.07890_F00209	017	—	\min \left\{\frac{f c}{w_{1}}, \frac{g c}{w_{2}}, 1\right\}	$\min \left\{\frac{f c}{w_{1}}, \frac{g c}{w_{2}}, 1\right\}$
2106.07890_F00210	017	—	c \mapsto \min \left\{\frac{f c}{w_{1}}, \frac{g c}{w_{2}}, 1\right\}	$c \mapsto \min \left\{\frac{f c}{w_{1}}, \frac{g c}{w_{2}}, 1\right\}$
2106.07890_F00211	017	—	Z: \mathcal{L} \rightarrow[0,1]	$Z: \mathcal{L} \rightarrow[0,1]$
2106.07890_F00212	017	—	\widehat{\mathcal{L}}(Z, F)	$\widehat{\mathcal{L}}(Z, F)$
2106.07890_F00213	017	—	J \rightarrow[0,1]	$J \rightarrow[0,1]$
2106.07890_F00214	017	—	J	J

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Identifier	Page	Conf.	LaTeX source	Rendered
2106.07890_F00215	017	—	[0,1]^{\mathcal{J}}	$[0,1]^{\mathcal{J}}$
2106.07890_F00216	017	—	f=F(1)	$f=F(1)$
2106.07890_F00217	017	—	g=F(2)	$g=F(2)$
2106.07890_F00218	018	—	\mathcal{L}(c, d) \leq\left[\lim _{W} F(c), \lim _{W} F(d)\right]	$\mathcal{L}(c, d) \leq\left[\lim _{W} F(c), \lim _{W} F(d)\right]$
2106.07890_F00219	018	—	\mathcal{L}(c, d) \lim _{W} F(c) \leq	$\mathcal{L}(c, d) \lim _{W} F(c) \leq$
2106.07890_F00220	018	—	\lim _{W} F(d)	$\lim _{W} F(d)$
2106.07890_F00221	018	—	c, d \in \mathcal{L}	$c, d \in \mathcal{L}$
2106.07890_F00222	018	—	g	g
2106.07890_F00223	018	—	\left(w_{1}, f\right) \times\left(w_{2}, g\right)	$\left(w_{1}, f\right) \times\left(w_{2}, g\right)$
2106.07890_F00224	018	—	w_{1}, w_{2} \in[0,1]	$w_{1}, w_{2} \in[0,1]$
2106.07890_F00225	018	—	\left(w_{1}, h^{x}\right) \times\left(w_{2}, h^{y}\right): \mathcal{L} \rightarrow[0,1]	$\left(w_{1}, h^{x}\right) \times\left(w_{2}, h^{y}\right): \mathcal{L} \rightarrow[0,1]$
2106.07890_F00226	018	—	\pi(c \mid x) \leq w_{1}	$\pi(c \mid x) \leq w_{1}$
2106.07890_F00227	018	—	\pi(c \mid y) \leq w_{2}	$\pi(c \mid y) \leq w_{2}$
2106.07890_F00228	018	—	w_{1} h^{x}(c) \times w_{2} h^{y}(c)	$w_{1} h^{x}(c) \times w_{2} h^{y}(c)$
2106.07890_F00229	018	—	\left(w_{1}, h^{x}\right) \times\left(w_{2}, h^{y}\right)	$\left(w_{1}, h^{x}\right) \times\left(w_{2}, h^{y}\right)$
2106.07890_F00230	018	—	w_{1}=w_{2}=1	$w_{1}=w_{2}=1$
2106.07890_F00231	018	—	f \times g	$f \times g$
2106.07890_F00232	018	—	f, g, h: \mathcal{L} \rightarrow[0,1]	$f, g, h: \mathcal{L} \rightarrow[0,1]$
2106.07890_F00233	019	—	W: \mathcal{J}^{\mathrm{op}} \rightarrow[0,1]	$W: \mathcal{J}^{\mathrm{op}} \rightarrow[0,1]$
2106.07890_F00234	019	—	w_{2}:=W(2)	$w_{2}:=W(2)$
2106.07890_F00235	019	—	F: \mathcal{J} \rightarrow \widehat{\mathcal{L}}	$F: \mathcal{J} \rightarrow \widehat{\mathcal{L}}$
2106.07890_F00236	019	—	\left(w_{1}, f\right) \sqcup\left(w_{2}, g\right): \mathcal{L} \rightarrow[0,1]	$\left(w_{1}, f\right) \sqcup\left(w_{2}, g\right): \mathcal{L} \rightarrow[0,1]$
2106.07890_F00237	019	—	\max \left\{w_{1} f c, w_{2} g c\right\}	$\max \left\{w_{1} f c, w_{2} g c\right\}$
2106.07890_F00238	019	—	\mathcal{J}=\mathcal{J}^{\mathrm{op}}	$\mathcal{J}=\mathcal{J}^{\mathrm{op}}$
2106.07890_F00239	019	—	\widehat{\mathcal{L}}(F-, Z)	$\widehat{\mathcal{L}}(F-, Z)$
2106.07890_F00240	019	—	i=1,2	$i=1,2$
2106.07890_F00241	020	—	c \mapsto \max \left\{w_{1} f c, w_{2} g c\right\}	$c \mapsto \max \left\{w_{1} f c, w_{2} g c\right\}$
2106.07890_F00242	020	—	\mathcal{L}(c, d) \leq\left[\operatorname{colim}_{W} F(c), \operatorname{colim}_{W} F(d)\right]	$\mathcal{L}(c, d) \leq\left[\operatorname{colim}_{W} F(c), \operatorname{colim}_{W} F(d)\right]$
2106.07890_F00243	020	—	\mathcal{L}(c, d) \operatorname{colim}_{W} F(c) \leq \operatorname{colim}_{W} F(d)	$\mathcal{L}(c, d) \operatorname{colim}_{W} F(c) \leq \operatorname{colim}_{W} F(d)$
2106.07890_F00244	020	—	\left(w_{1}, f\right) \sqcup\left(w_{2}, g\right)	$\left(w_{1}, f\right) \sqcup\left(w_{2}, g\right)$
2106.07890_F00245	020	—	G, H: \mathrm{C} \rightarrow	$G, H: \mathrm{C} \rightarrow$
2106.07890_F00246	020	—	c \mapsto \widehat{\mathrm{C}}\left(h^{c} \times G, H\right)	$c \mapsto \widehat{\mathrm{C}}\left(h^{c} \times G, H\right)$
2106.07890_F00247	020	—	h^{c} \times G	$h^{c} \times G$
2106.07890_F00248	020	—	[f, g]: \mathcal{L} \rightarrow[0,1]	$[f, g]: \mathcal{L} \rightarrow[0,1]$
2106.07890_F00249	020	—	h^{c} \times f	$h^{c} \times f$
2106.07890_F00250	020	—	\mathcal{L}(c, d) \leq[[f, g](c),[f, g](d)]	$\mathcal{L}(c, d) \leq[[f, g](c),[f, g](d)]$
2106.07890_F00251	020	—	\mathcal{L}(c, d)[f, g](c) \leq[f, g](d)	$\mathcal{L}(c, d)[f, g](c) \leq[f, g](d)$
2106.07890_F00252	020	—	\mathcal{L}(c, d) \leq[g(c), g(d)]	$\mathcal{L}(c, d) \leq[g(c), g(d)]$
2106.07890_F00253	020	—	\mathcal{L}(c, d) g(c) \leq g(d)	$\mathcal{L}(c, d) g(c) \leq g(d)$
2106.07890_F00254	020	—	\mathcal{L}(c, d) \widehat{\mathcal{L}}\left(h^{c}, g\right) \leq \widehat{\mathcal{L}}\left(h^{d}, g\right)	$\mathcal{L}(c, d) \widehat{\mathcal{L}}\left(h^{c}, g\right) \leq \widehat{\mathcal{L}}\left(h^{d}, g\right)$
2106.07890_F00255	020	—	\mathcal{L}(c, d)\left(\widehat{\mathcal{L}}\left(h^{c}, g\right) \times \widehat{\mathcal{L}}(f, g\right) \leq \widehat{\mathcal{L}}\left(h^{d}, g\right) \times \widehat{\mathcal{L}}(f, g)	$\mathcal{L}(c, d)\left(\widehat{\mathcal{L}}\left(h^{c}, g\right) \times \widehat{\mathcal{L}}(f, g\right) \leq \widehat{\mathcal{L}}\left(h^{d}, g\right) \times \widehat{\mathcal{L}}(f, g)$
2106.07890_F00256	020	—	\mathcal{L}(c, d) \widehat{\mathcal{L}}\left(h^{c} \times f, g\right) \leq \widehat{\mathcal{L}}\left(h^{d} \times f, g\right)	$\mathcal{L}(c, d) \widehat{\mathcal{L}}\left(h^{c} \times f, g\right) \leq \widehat{\mathcal{L}}\left(h^{d} \times f, g\right)$
2106.07890_F00257	021	—	\mathcal{L}(c, d) \leq\left[\widehat{\mathcal{L}}\left(h^{c} \times f, g\right), \widehat{ \mathcal{L}}\left(h^{d} \times f, g\right)\right]=[[f, g](c),[f, g](d)]	$\mathcal{L}(c, d) \leq\left[\widehat{\mathcal{L}}\left(h^{c} \times f, g\right), \widehat{\mathcal{L}}\left(h^{d} \times f, g\right)\right]=[[f, g](c),[f, g](d)]$
2106.07890_F00258	021	—	(f \times-): \widehat{\mathcal{L}} \rightarrow \widehat{\mathcal{L}}	$(f \times-): \widehat{\mathcal{L}} \rightarrow \widehat{\mathcal{L}}$
2106.07890_F00259	021	—	[f,-]	$[f,-]$
2106.07890_F00260	021	—	-\times f	$-\times f$
2106.07890_F00261	021	—	\widehat{\mathcal{L}}\left(h^{c},[f, g]\right)=[f, g](c)	$\widehat{\mathcal{L}}\left(h^{c},[f, g]\right)=[f, g](c)$
2106.07890_F00262	021	—	[f, g](c)=\widehat{\mathcal{L}}\left(h^{c} \times f, g\right)	$[f, g](c)=\widehat{\mathcal{L}}\left(h^{c} \times f, g\right)$
2106.07890_F00263	021	—	\widehat{\mathcal{L}}\left(h^{c},[f, g]\right)=\widehat{\mathcal{L}}\left(h^{c} \times f, g\right)	$\widehat{\mathcal{L}}\left(h^{c},[f, g]\right)=\widehat{\mathcal{L}}\left(h^{c} \times f, g\right)$
2106.07890_F00264	021	—	h^{c}	h^{c}
2106.07890_F00265	021	—	h	h

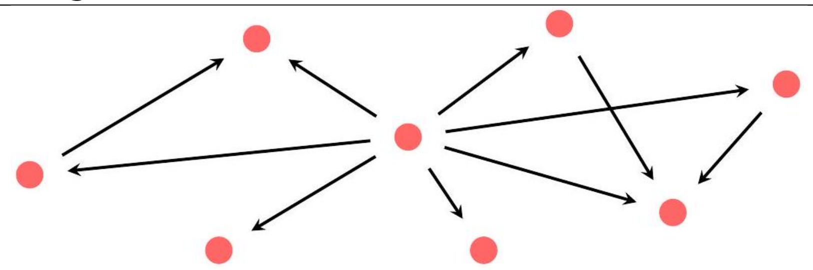
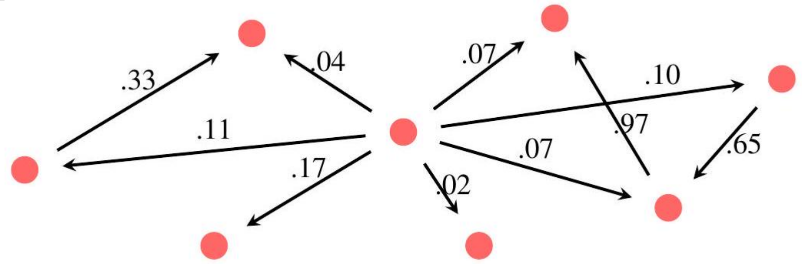
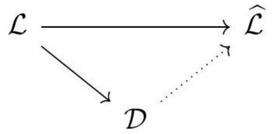
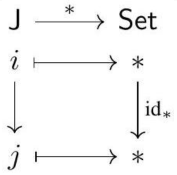
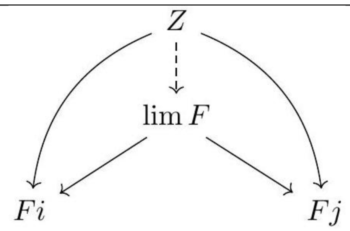
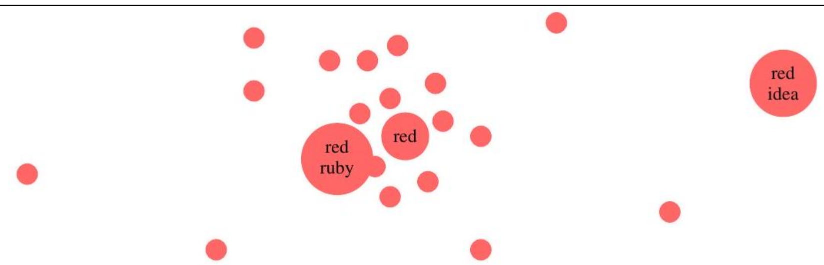
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Identifier	Page	Conf.	LaTeX source	Rendered
2106.07890_F00266	021	—	`[f, g]`	$[f, g]$
2106.07890_F00267	021	—	`x, y \in \mathcal{L}`	$x, y \in \mathcal{L}$
2106.07890_F00268	021	—	`\left[h^{x}, h^{y}\right](c)=`	$\left[h^{x}, h^{y}\right](c)=$
2106.07890_F00269	021	—	`d=c`	$d=c$
2106.07890_F00270	021	—	`\pi(c \mid y)=`	$\pi(c \mid y)=$
2106.07890_F00271	021	—	`\pi(c \mid c) \times \pi(c \mid x)=\pi(c \mid x) \neq 0`	$\pi(c \mid c) \times \pi(c \mid x)=\pi(c \mid x) \neq 0$
2106.07890_F00272	021	—	`\left[h^{x}, h^{y}\right](c) \neq 0`	$\left[h^{x}, h^{y}\right](c) \neq 0$
2106.07890_F00273	021	—	`x \Rightarrow y`	$x \Rightarrow y$
2106.07890_F00274	021	—	`\left[h^{x}, h^{y}\right]: \mathcal{L} \rightarrow[0,1]`	$\left[h^{x}, h^{y}\right]: \mathcal{L} \rightarrow[0,1]$
2106.07890_F00275	021	—	`\mathcal{M}`	$\mathcal{M}$
2106.07890_F00276	021	—	`[0, \infty]`	$[0, \infty]$
2106.07890_F00277	021	—	`a+\infty:=\infty`	$a+\infty:=\infty$
2106.07890_F00278	021	—	`\infty+a:=\infty`	$\infty+a:=\infty$
2106.07890_F00279	021	—	`b \leq a`	$b \leq a$
2106.07890_F00280	022	—	`d_{\mathcal{M}}`	$d_{\mathcal{M}}$
2106.07890_F00281	022	—	`[a, b]:=\max \{b-a, 0\}`	$[a, b]:=\max \{b-a, 0\}$
2106.07890_F00282	022	—	`a \times b=\max \{a, b\}`	$a \times b=\max \{a, b\}$
2106.07890_F00283	022	—	`a \sqcup b=\min \{a, b\}`	$a \sqcup b=\min \{a, b\}$
2106.07890_F00284	022	—	`-\ln :[0,1] \rightarrow[0, \infty]`	$-\ln :[0,1] \rightarrow[0, \infty]$
2106.07890_F00285	022	—	`[0, \infty] \rightarrow[0,1]`	$[0, \infty] \rightarrow[0,1]$
2106.07890_F00286	022	—	`a \mapsto \exp (-a)`	$a \mapsto \exp (-a)$
2106.07890_F00287	022	—	`\mathcal{M}(x, y):=-\ln \mathcal{L}(x, y)`	$\mathcal{M}(x, y):=-\ln \mathcal{L}(x, y)$
2106.07890_F00288	022	—	`0 \geq \mathcal{M}(x, x)`	$0 \geq \mathcal{M}(x, x)$
2106.07890_F00289	022	—	`\mathcal{M}(x, y)+\mathcal{M}(y, z) \geq \mathcal{M}(x, z)`	$\mathcal{M}(x, y)+\mathcal{M}(y, z) \geq \mathcal{M}(x, z)$
2106.07890_F00290	022	—	`d_{\mathcal{M}}(x, y):=\mathcal{M}(x, y)`	$d_{\mathcal{M}}(x, y):=\mathcal{M}(x, y)$
2106.07890_F00291	022	—	`\widehat{\mathcal{M}}:=[0, \infty]^{\mathcal{M}}`	$\widehat{\mathcal{M}}:=[0, \infty]^{\mathcal{M}}$
2106.07890_F00292	023	—	`f: \mathcal{M} \rightarrow[0, \infty]`	$f: \mathcal{M} \rightarrow[0, \infty]$
2106.07890_F00293	023	—	`d_{M}(x, y) \geq[f x, f y]=\max \{f y-f x, 0\}`	$d_{M}(x, y) \geq[f x, f y]=\max \{f y-f x, 0\}$
2106.07890_F00294	023	—	`[a, b]=\max \{b-a, 0\}`	$[a, b]=\max \{b-a, 0\}$
2106.07890_F00295	023	—	`d_{[0, \infty]}(a, b)`	$d_{[0, \infty]}(a, b)$
2106.07890_F00296	023	—	`d_{[0, \infty]}(f x, f y) \leq d_{M}(x, y)`	$d_{[0, \infty]}(f x, f y) \leq d_{M}(x, y)$
2106.07890_F00297	023	—	`\widehat{\mathcal{M}}`	$\widehat{\mathcal{M}}$
2106.07890_F00298	023	—	`\mathcal{M}^{\mathrm{op}} \rightarrow \widehat{\mathcal{M}}`	$\mathcal{M}^{\mathrm{op}} \rightarrow \widehat{\mathcal{M}}$
2106.07890_F00299	023	—	`x \mapsto d_{\mathcal{M}}(x,-)`	$x \mapsto d_{\mathcal{M}}(x,-)$
2106.07890_F00300	023	—	`f, g \in \widehat{\mathcal{M}}`	$f, g \in \widehat{\mathcal{M}}$
2106.07890_F00301	023	—	`w_{1}, w_{2}`	w_{1}, w_{2}
2106.07890_F00302	023	—	`f, g: \mathcal{M} \rightarrow[0, \infty]`	$f, g: \mathcal{M} \rightarrow[0, \infty]$
2106.07890_F00303	023	—	`w_{1}, w_{2} \in[0, \infty]`	$w_{1}, w_{2} \in[0, \infty]$
2106.07890_F00304	023	—	`f^{\prime}, g^{\prime}: \mathcal{L} \rightarrow[0,1]`	$f^{\prime}, g^{\prime}: \mathcal{L} \rightarrow[0,1]$
2106.07890_F00305	023	—	`w_{1}^{\prime}, w_{2}^{\prime} \in[0,1]`	$w_{1}^{\prime}, w_{2}^{\prime} \in[0,1]$
2106.07890_F00306	023	—	`f^{\prime} c=\exp (-f c), g^{\prime} c=\exp (-g c)`	$f^{\prime} c=\exp (-f c), g^{\prime} c=\exp (-g c)$
2106.07890_F00307	023	—	`w_{i}^{\prime}=\exp \left(-w_{i}\right)`	$w_{i}^{\prime}=\exp \left(-w_{i}\right)$
2106.07890_F00308	024	—	`d_{\mathcal{M}}(x,-)`	$d_{\mathcal{M}}(x,-)$
2106.07890_F00309	024	—	`((-\infty, \infty], \oplus, \odot)`	$((-\infty, \infty], \oplus, \odot)$
2106.07890_F00310	024	—	`([0, \infty], \oplus, \odot)`	$([0, \infty], \oplus, \odot)$
2106.07890_F00311	024	—	`w_{1}=w_{2}=0`	$w_{1}=w_{2}=0$
2106.07890_F00312	024	—	`[0, \infty] \times \widehat{\mathcal{M}} \rightarrow \widehat{\mathcal{M}}`	$[0, \infty] \times \widehat{\mathcal{M}} \rightarrow \widehat{\mathcal{M}}$
2106.07890_F00313	024	—	`(s, f) \mapsto s \odot f`	$(s, f) \mapsto s \odot f$
2106.07890_F00314	024	—	`(s \cdot f)(x)=f(x)+s`	$(s \cdot f)(x)=f(x)+s$
2106.07890_F00315	024	—	`\left(s_{1} \oplus s_{2}\right) f(x)=\min \left\{s_{1}, s_{2}\right\} f(x)=\min \left\{f x+s_{1}, f x+s_{2}\right\}`	$\left(s_{1} \oplus s_{2}\right) f(x)=\min \left\{s_{1}, s_{2}\right\} f(x)=\min \left\{f x+s_{1}, f x+s_{2}\right\}$
2106.07890_F00316	024	—	`\left(s_{1} \odot s_{2}\right) f(x)=f x+s_{1}+s_{2}`	$\left(s_{1} \odot s_{2}\right) f(x)=f x+s_{1}+s_{2}$
2106.07890_F00317	024	—	`\left(s_{1}\right) \odot\left(s_{2} \odot f\right)(x)=`	$\left(s_{1}\right) \odot\left(s_{2} \odot f\right)(x)=$
2106.07890_F00318	024	—	`\left(s_{2} \odot f\right)(x)+s_{1}=f x+s_{2}+s_{1}`	$\left(s_{2} \odot f\right)(x)+s_{1}=f x+s_{2}+s_{1}$
2106.07890_F00319	026	—	`\sqcup`	$\sqcup$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Unrecovered image regions — TikZ / table / failed-math candidates

Regions MathPix left as images (no LaTeX). Reconstruct with `pdfdrill vision` (LLM classify + transcribe to TikZ/tabular/math); verify an LLM result against the real ink with inkdrill.

Identifier	Page	Image	Source
2106.07890_DIA_0001	005		tex.zip (local)
2106.07890_DIA_0002	009		tex.zip (local)
2106.07890_DIA_0003	012		tex.zip (local)
2106.07890_DIA_0004	016		tex.zip (local)
2106.07890_DIA_0005	016		tex.zip (local)
2106.07890_DIA_0006	022		tex.zip (local)